

R3 Understanding weight

Name: _____

Class: _____

Date: _____

Time: **373 minutes**

Marks: **313 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$

A ball, initially at rest, is dropped from a height of 40 m above the ground.

Calculate the speed of the ball when it reaches the ground.

Circle your answer.

-28 m s^{-1}

28 m s^{-1}

-780 m s^{-1}

780 m s^{-1}

(Total 1 mark)

Q2.

A remote-controlled toy car is moving over a horizontal surface. It moves in a straight line through a point A.

The toy is initially at the point with displacement 3 metres from A. Its velocity, $v \text{ m s}^{-1}$, at time t seconds is defined by

$$v = 0.06(2 + t - t^2)$$

- (a) Find an expression for the displacement, r metres, of the toy from A at time t seconds.

(4)

- (b) **In this question use $g = 9.8 \text{ m s}^{-2}$**

At time $t = 2$ seconds, the toy launches a ball which travels directly upwards with initial speed 3.43 m s^{-1}

Find the time taken for the ball to reach its highest point.

(3)

(Total 7 marks)

Q3.

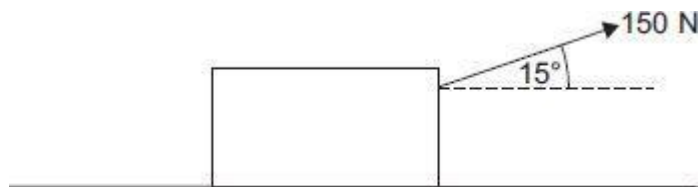
In this question use $g = 9.8 \text{ m s}^{-2}$

A boy attempts to move a wooden crate of mass 20 kg along horizontal ground. The coefficient of friction between the crate and the ground is 0.85

- (a) The boy applies a horizontal force of 150 N. Show that the crate remains stationary.

(3)

- (b) Instead, the boy uses a handle to pull the crate forward. He exerts a force of 150 N, at an angle of 15° above the horizontal, as shown in the diagram.



Determine whether the crate remains stationary.

Fully justify your answer.

(5)
(Total 8 marks)

Q4.

In this question use $g = 10 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A man of mass 80 kg is travelling in a lift.
The lift is rising vertically.



The lift decelerates at a rate of 1.5 m s^{-2}

Find the magnitude of the force exerted on the man by the lift.

(Total 3 marks)

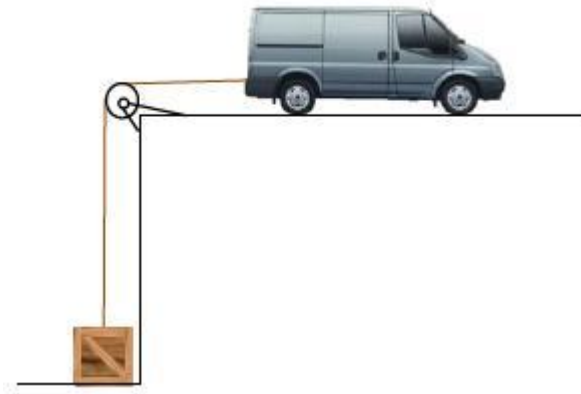
Q5.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A van of mass 1300 kg and a crate of mass 300 kg are connected by a light inextensible rope.

The rope passes over a light smooth pulley, as shown in the diagram.

The rope between the pulley and the van is horizontal.



Initially, the van is at rest and the crate rests on the lower level. The rope is taut. The van moves away from the pulley to lift the crate from the lower level. The van's engine produces a constant driving force of 5000 N. A constant resistance force of magnitude 780 N acts on the van. Assume there is no resistance force acting on the crate.

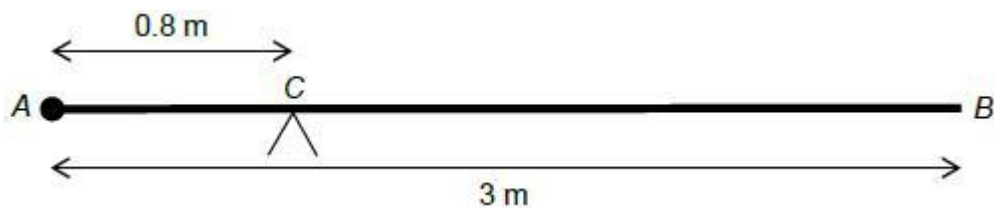
- (a) (i) Draw a diagram to show the forces acting on the crate while it is being lifted. (1)
- (ii) Draw a diagram to show the forces acting on the van while the crate is being lifted. (1)
- (b) Show that the acceleration of the van is 0.80 m s^{-2} (4)
- (c) Find the tension in the rope. (2)
- (d) Suggest how the assumption of a constant resistance force could be refined to produce a better model. (1)
- (Total 9 marks)

Q6.

A uniform rod, AB , has length 3 metres and mass 24 kg.

A particle of mass M kg is attached to the rod at A .

The rod is balanced in equilibrium on a support at C , which is 0.8 metres from A .



Find the value of M .

(Total 2 marks)

Q7.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally. The velocity, $\mathbf{v} \text{ m s}^{-1}$, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that the parachutist is at the origin when $t = 0$

Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t .

(4)

- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)

(Total 11 marks)

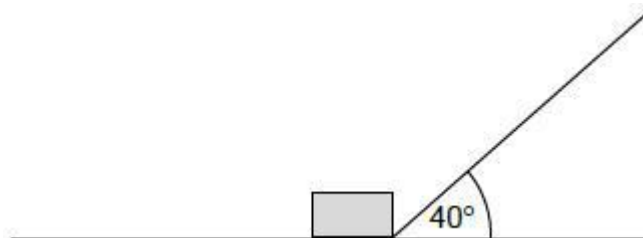
Q8.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg , being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons.



The coefficient of friction between the box and the board is μ

Model the box as a particle.

- (a) Show that $\mu = 0.83$

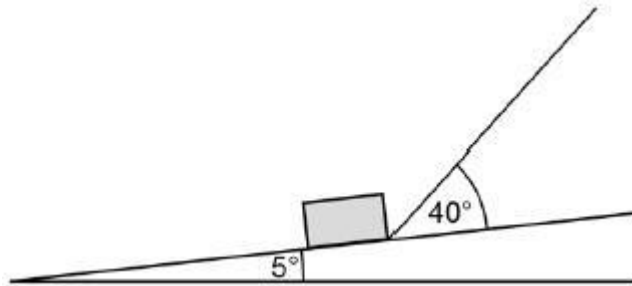
(4)

- (b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.

The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



- (i) Draw a diagram to show the forces acting on the box as it moves.

(1)

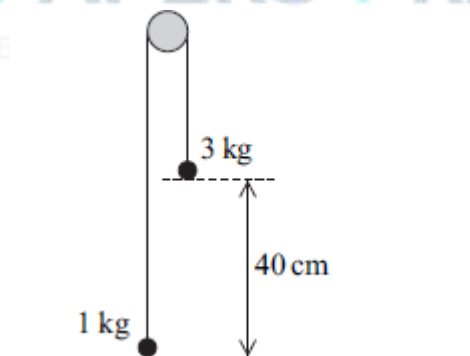
- (ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2} .

(7)

(Total 12 marks)

Q9.

Two particles are connected by a light inextensible string that passes over a smooth peg. The particles have masses of 3 kg and 1 kg . The 1 kg particle is pulled down to ground level, where it is 40 cm below the level of the 3 kg particle, as shown in the diagram.



The particles are released from rest with the string vertical above each particle. Assume that no resistance forces act on the particles as they move.

- (a) By forming two equations of motion, one for each particle, find the magnitude of the acceleration of the particles after they have been released but before the 3 kg particle hits the ground.

(5)

- (b) Find the speed of the 1 kg particle when the 3 kg particle hits the ground. (2)
- (c) After the 3 kg particle has hit the ground, the 1 kg particle continues to move and the string is now slack. Find the maximum height above ground level reached by the 1 kg particle. (3)
- (d) If a constant air resistance force also acts on the particles as they move, explain how this would change your answer for the acceleration in part (a). Give a reason for your answer. (2)
- (Total 12 marks)

Q10.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4.

- (a) The tension in the rope is 150 newtons.
- (i) Draw a diagram to show the forces acting on the block as it moves. (2)
- (ii) Show that the magnitude of the normal reaction force on the block is 243 newtons, correct to three significant figures. (3)
- (iii) Find the magnitude of the friction force acting on the block. (2)
- (iv) Find the acceleration of the block. (4)
- (b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed. (5)
- (c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)? (1)
- (Total 17 marks)

Q11.

A block, of mass 4 kg, is made to move in a straight line on a rough horizontal surface by a horizontal force of 50 newtons, as shown in the diagram.

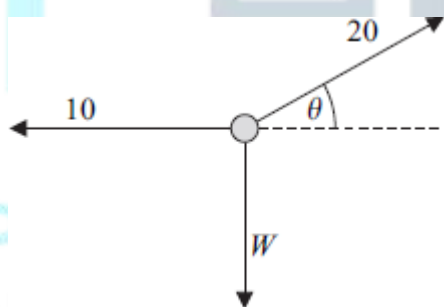


Assume that there is no air resistance acting on the block.

- (a) Draw a diagram to show all the forces acting on the block. (1)
 - (b) Find the magnitude of the normal reaction force acting on the block. (1)
 - (c) The acceleration of the block is 3 m s^{-2} . Find the magnitude of the friction force acting on the block. (3)
 - (d) Find the coefficient of friction between the block and the surface. (2)
 - (e) Explain how and why your answer to part (d) would change if you assumed that air resistance did act on the block. (2)
- (Total 9 marks)**

Q12.

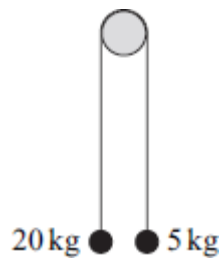
A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



- (a) Find θ . (3)
 - (b) Find W . (2)
 - (c) Calculate the mass of the particle. (2)
- (Total 7 marks)**

Q13.

Two particles have masses of 20 kg and 5 kg. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



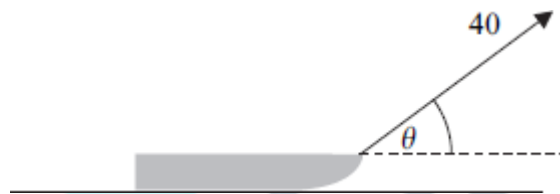
The particles are released from rest.

By forming two equations of motion, find the magnitude of the acceleration of the particles.

(Total 4 marks)

Q14.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of θ to the horizontal, as shown in the diagram.



Model the sledge as a particle.

- (a) Draw a diagram to show all the forces acting on the sledge. (1)
- (b) Find the magnitude of the normal reaction force acting on the sledge, in terms of θ . (3)
- (c) The acceleration of the sledge is

$$p + q \cos \theta + r \sin \theta$$

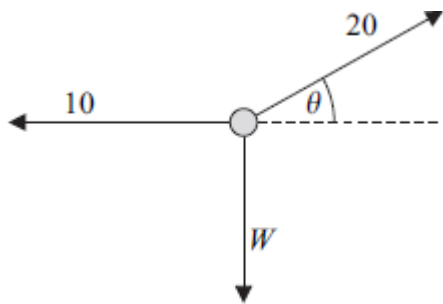
Find p , q and r .

(5)

(Total 9 marks)

Q15.

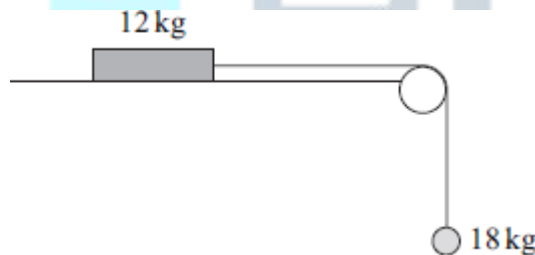
A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



- (a) Find θ . (3)
- (b) Find W . (2)
- (c) Calculate the mass of the particle. (2)
- (Total 7 marks)**

Q16.

A block, of mass 12 kg, lies on a horizontal surface. The block is attached to a particle, of mass 18 kg, by a light inextensible string which passes over a smooth fixed peg. Initially, the block is held at rest so that the string supports the particle, as shown in the diagram.



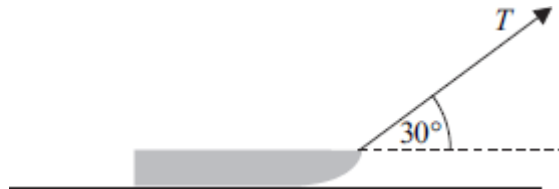
The block is then released.

- (a) Assuming that the surface is smooth, use two equations of motion to find the magnitude of the acceleration of the block and particle. (4)
- (b) In reality, the surface is rough and the acceleration of the block is 3 m s^{-2} . (3)
- (i) Find the tension in the string. (1)
- (ii) Calculate the magnitude of the normal reaction force acting on the block. (5)
- (iii) Find the coefficient of friction between the block and the surface. (5)
- (c) State two modelling assumptions, other than those given, that you have made in answering this question.

(2)
(Total 15 marks)

Q17.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is T newtons. The rope is kept at an angle of 30° to the horizontal, as shown in the diagram.



Model the sledge as a particle.

- (a) Draw a diagram to show all the forces acting on the sledge. (1)
 - (b) Find the magnitude of the normal reaction force acting on the sledge, in terms of T . (3)
 - (c) Given that the sledge accelerates at 0.05 m s^{-2} , find T . (6)
- (Total 10 marks)

Q18.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s^{-1} .

At time t seconds after it is projected, the stone has speed $v \text{ m s}^{-1}$ and it experiences a resistance force of magnitude $9.8v$ newtons.

- (a) When $t \geq 0$, show that

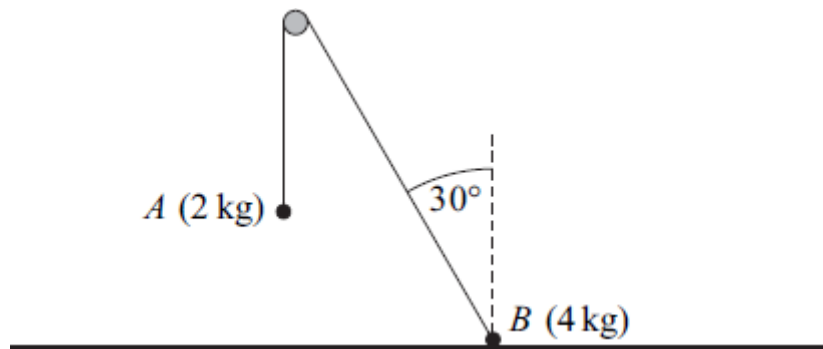
$$\frac{dv}{dt} = -1.96(v - 5) \quad (2)$$

- (b) Find v in terms of t . (5)
- (Total 7 marks)

Q19.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth peg. Particle A has mass 2 kg and particle B has mass 4 kg. Particle A hangs freely with the string vertical. Particle B is at rest in equilibrium on a rough horizontal

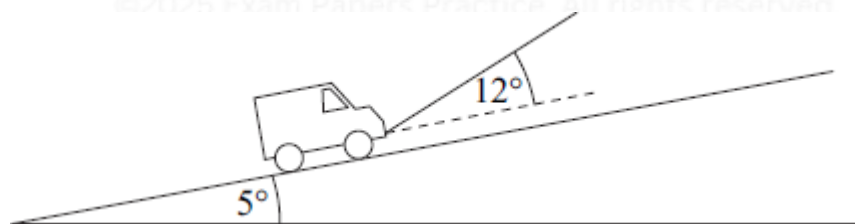
surface with the string at an angle of 30° to the vertical. The particles, peg and string are shown in the diagram.



- (a) By considering particle A , find the tension in the string. (2)
 - (b) Draw a diagram to show the forces acting on particle B . (2)
 - (c) Show that the magnitude of the normal reaction force acting on particle B is 22.2 newtons, correct to three significant figures. (3)
 - (d) Find the least possible value of the coefficient of friction between particle B and the surface. (4)
- (Total 11 marks)**

Q20.

A van, of mass 2000 kg, is towed up a slope inclined at 5° to the horizontal. The tow rope is at an angle of 12° to the slope. The motion of the van is opposed by a resistance force of magnitude 500 newtons. The van is accelerating up the slope at 0.6 m s^{-2} .



Model the van as a particle.

- (a) Draw a diagram to show the forces acting on the van. (2)
 - (b) Show that the tension in the tow rope is 3480 newtons, correct to three significant figures. (5)
- (Total 7 marks)**

Q21.

A crane is used to lift a load, using a single vertical cable which is attached to the load. The load accelerates uniformly from rest. When it has risen 0.9 metres, its speed is 0.6 m s^{-1} .

- (a) (i) Show that the acceleration of the load is 0.2 m s^{-2} . (3)
- (ii) Find the time taken for the load to rise 0.9 metres. (2)
- (b) Given that the mass of the load is 800 kg, find the tension in the cable while the load is accelerating. (3)

(Total 8 marks)

Q22.

A wooden block, of mass 4 kg, is placed on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.3. A horizontal force, of magnitude 30 newtons, acts on the block and causes it to accelerate.

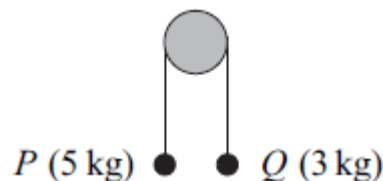


- (a) Draw a diagram to show all the forces acting on the block. (1)
- (b) Calculate the magnitude of the normal reaction force acting on the block. (1)
- (c) Find the magnitude of the friction force acting on the block. (2)
- (d) Find the acceleration of the block. (3)

(Total 7 marks)

Q23.

Two particles, P and Q , are connected by a string that passes over a fixed smooth peg, as shown in the diagram. The mass of P is 5 kg and the mass of Q is 3 kg.

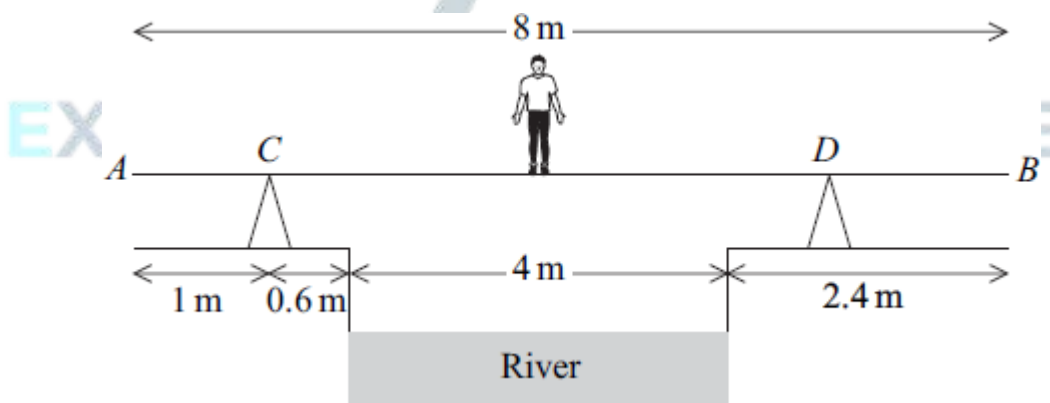


The particles are released from rest in the position shown.

- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.45 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) State **two** modelling assumptions that you have made about the string. (2)
- (d) Particle P hits the floor when it has moved 0.196 metres and Q has not reached the peg.
- (i) Find the time that it takes P to reach the floor. (3)
- (ii) Find the speed of P when it hits the floor. (2)
- (Total 14 marks)

Q24.

Ken is trying to cross a river of width 4 m . He has a uniform plank, AB , of length 8 m and mass 17 kg . The ground on both edges of the river bank is horizontal. The plank rests at two points, C and D , on fixed supports which are on opposite sides of the river. The plank is at right angles to both river banks and is horizontal. The distance AC is 1 m , and the point C is at a horizontal distance of 0.6 m from the river bank. Ken, who has mass 65 kg , stands on the plank directly above the middle of the river, as shown in the diagram.



- (a) Draw a diagram to show the forces acting on the plank. (2)
- (b) Given that the reaction on the plank at the point D is $44g \text{ N}$, find the horizontal distance of the point D from the nearest river bank. (4)
- (c) State how you have used the fact that the plank is uniform in your solution. (1)
- (Total 7 marks)

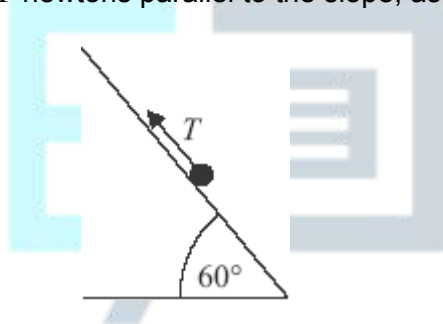
Q25.

A ball is thrown vertically upwards at a speed of 4.9 m s^{-1} from a height of 1 metre above ground level. The ball is caught when it returns to a height of 1 metre above ground level and is travelling downwards.

- (a) Find the time that it takes for the ball to reach its maximum height. (3)
 - (b) Find the maximum height of the ball above ground level. (3)
 - (c) State the total time for which the ball is moving. (1)
 - (d) State the speed of the ball when it is caught. (1)
- (Total 8 marks)**

Q26.

A particle of mass 3 kg is on a smooth slope inclined at 60° to the horizontal. The particle is held at rest by a force of T newtons parallel to the slope, as shown in the diagram.



- (a) Draw a diagram to show all the forces acting on the particle. (1)
 - (b) Show that the magnitude of the normal reaction acting on the particle is 14.7 newtons. (2)
 - (c) Find T . (2)
- (Total 5 marks)**

Q27.

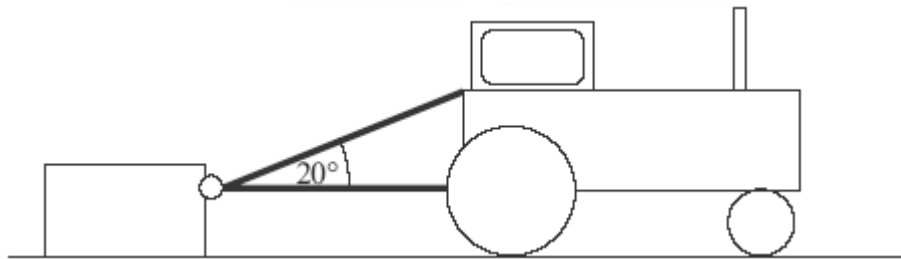
A ball is released from rest at a height of 15 metres above ground level.

- (a) Find the speed of the ball when it hits the ground, assuming that no air resistance acts on the ball. (3)
- (b) In fact, air resistance does act on the ball. Assume that the air resistance force has a constant magnitude of 0.9 newtons. The ball has a mass of 0.5 kg.

- (i) Draw a diagram to show the forces acting on the ball, including the magnitudes of the forces acting. (1)
 - (ii) Show that the acceleration of the ball is 8 m s^{-2} . (3)
 - (iii) Find the speed at which the ball hits the ground. (2)
 - (iv) Explain why the assumption that the air resistance force is constant may not be valid. (1)
- (Total 10 marks)**

Q28.

A crate, of mass 200 kg , is initially at rest on a rough horizontal surface. A smooth ring is attached to the crate. A light inextensible rope is passed through the ring, and each end of the rope is attached to a tractor. The lower part of the rope is horizontal and the upper part is at an angle of 20° to the horizontal, as shown in the diagram.



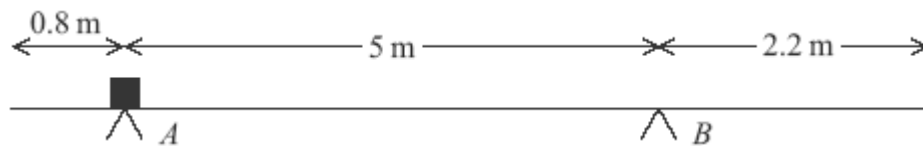
When the tractor moves forward, the crate accelerates at 0.3 m s^{-2} . The coefficient of friction between the crate and the surface is 0.4 .

Assume that the tension, T newtons, is the same in both parts of the rope.

- (a) Draw and label a diagram to show the forces acting on the crate. (2)
 - (b) Express the normal reaction between the surface and the crate in terms of T . (3)
 - (c) Find T . (5)
- (Total 10 marks)**

Q29.

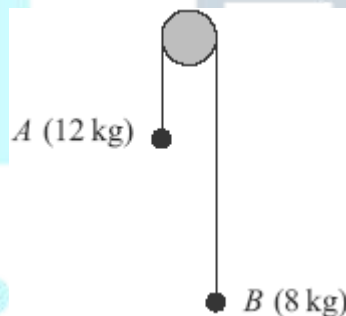
A uniform plank, of length 8 metres , has mass 30 kg . The plank is supported in equilibrium in a horizontal position by two smooth supports at the points A and B , as shown in the diagram. A block, of mass 20 kg , is placed on the plank at point A .



- (a) Draw a diagram to show the forces acting on the plank. (2)
 - (b) Show that the magnitude of the force exerted on the plank by the support at B is $19.2g$ newtons. (3)
 - (c) Find the magnitude of the force exerted on the plank by the support at A . (2)
 - (d) Explain how you have used the fact that the plank is uniform in your solution. (1)
- (Total 8 marks)**

Q30.

Two particles, A and B , have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.
 - (i) Find the speed of particle A when the string breaks. (2)
 - (ii) Find the speed of particle A when it hits the surface.

(3)
(Total 12 marks)

Q31.

A block, of mass 10 kg, is at rest on a rough horizontal surface, when a horizontal force, of magnitude P newtons, is applied to the block, as shown in the diagram.

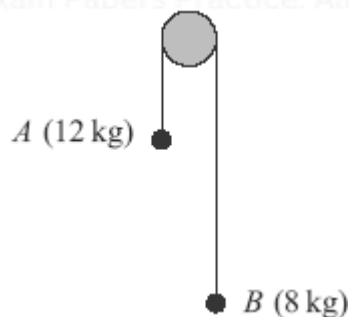


The coefficient of friction between the block and the surface is 0.5.

- (a) Draw and label a diagram to show all the forces acting on the block. **(1)**
 - (b) (i) Calculate the magnitude of the normal reaction force acting on the block. **(1)**
 - (ii) Find the maximum possible magnitude of the friction force between the block and the surface. **(1)**
 - (iii) Given that $P = 30$, state the magnitude of the friction force acting on the block. **(1)**
 - (c) Given that $P = 80$, find the acceleration of the block. **(3)**
- (Total 7 marks)**

Q32.

Two particles, A and B , have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



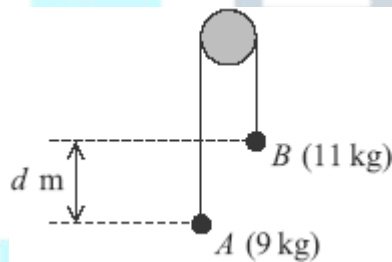
The particles are released from rest and move vertically. Assume that there is no air resistance.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2} . **(5)**

- (b) Find the tension in the string. (2)
- (c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.
- (i) Find the speed of particle A when the string breaks. (2)
- (ii) Find the speed of particle A when it hits the surface. (3)
- (iii) Find the time that it takes for particle B to reach the surface after the string breaks.
 Assume that particle B does not hit the peg. (5)
- (Total 17 marks)**

Q33.

Two particles, A and B , are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A .

Assume that no resistance forces act on the particles.

- (a) By forming an equation of motion for each of the particles A and B , show that the acceleration of each particle has magnitude 0.98 m s^{-2} . (5)
- (b) When the particles have been moving for 0.5 seconds, they are at the same level.
- (i) Find the speed of the particles at this time. (2)
- (ii) Find d . (4)
- (Total 11 marks)**

Q34.

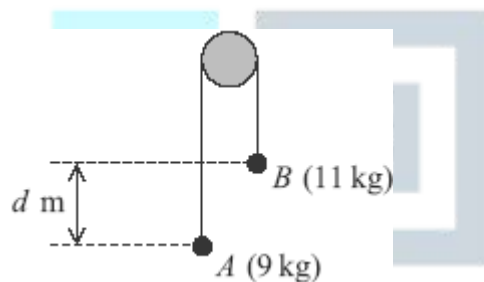
A box of mass 4 kg is held at rest on a plane inclined at an angle of 40° to the horizontal. The box is then released and slides down the plane.

- (a) A simple model assumes that the only forces acting on the box are its weight and the normal reaction from the plane. Show that, according to this simple model, the acceleration of the box would be 6.30 m s^{-2} , correct to three significant figures. (3)
- (b) In fact, the box moves down the plane with constant acceleration and travels 0.9 metres in 0.6 seconds. By using this information, find the acceleration of the box. (3)
- (c) Explain why the answer to part (b) is less than the answer to part (a). (1)

(Total 7 marks)

Q35.

Two particles, A and B , are connected by a string that passes over a fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A .

The motion of the particles is to be modelled using simple assumptions.

- (a) State one assumption that should be made about the peg. (1)
- (b) State two assumptions that should be made about the string. (2)
- (c) By forming an equation of motion for each of the particles A and B , show that the acceleration of each particle has magnitude 0.98 m s^{-2} . (5)
- (d) When the particles have been moving for 0.5 seconds, they are at the same level.
 - (i) Find the speed of the particles at this time. (2)
 - (ii) Find d . (4)

(Total 14 marks)



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